

Chaitanyaa Chopraa

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Summary

Backend-focused Software Engineer graduated Dec 2025 (University of Victoria) with experience building scalable services using Python, Node.js, REST APIs, and Docker. Strong foundation in distributed systems, testing, and production reliability, with hands-on experience designing microservices, message-driven architectures, and real-world hardware-integrated systems. Passionate about building reliable infrastructure that operates at scale.

Education

University of Victoria

BSc, Computer Science — Software Systems • Business Minor

Graduating GPA: 8.3 / 9.0

Victoria, BC, Canada

Sept 2021 – Dec 2025

Relevant Coursework: Software Engineering, Web Development, Databases, Algorithms and Data Structures, Operating Systems, Software Testing and QA

Skills

Languages: Python, JavaScript, TypeScript, Java, C/C++

Backend: Node.js, Express, REST APIs, Authentication (JWT)

DevOps & Systems: Docker, Linux, Git, CI/CD concepts

Distributed Systems: RabbitMQ, Redis, Microservices, Caching

Databases: MongoDB, PostgreSQL, MySQL

Testing & Quality: pytest, JUnit, JMeter, debugging, performance testing

Frontend: Soft Skills: Problem Solving, Systems Thinking, Ownership, Collaboration, Communication

Professional Experience

Houle Electric

Co-op Project Manager / Assistant

Jan 2024 – Aug 2024

Cowichan District Hospital Redevelopment

- Developed and implemented robust Quality Assurance (QA) and Quality Control (QC) processes for the Cowichan District Hospital redevelopment project.
- Successfully executed effective QA/QC workflows to ensure project completion on time and in compliance with quality, safety, and regulatory standards.
- Played a pivotal role in configuring Procore by designing a reusable, company-wide QA/QC template, now actively adopted across multiple projects.
- Managed high-volume documentation, inspections, and reporting, maintaining accuracy and accountability across field and management teams.

Infinity Outdoor Pvt. Ltd

Contract Web Developer

Delivered 2025

- Designed and launched a responsive marketing website to enhance brand visibility and showcase advertising campaigns.
- Built the application using React, Vite, and Tailwind CSS, achieving a 35% improvement in load times through optimized component architecture.
- Collaborated closely with stakeholders to align design, content, and functionality, delivering a modern and data-driven user experience.

Technical Projects

Social Spark — AI Brand & Influencer Platform

React, Node.js, MongoDB, Gemini API, Docker

- Developed an AI-driven platform to improve brand-influencer collaboration through intelligent campaign matching and analytics.
- Integrated the Gemini API to automate content evaluation, generate campaign recommendations, and provide engagement insights.
- Designed secure authentication and backend services to support scalable user interactions and data processing.
- Designed scalable backend services with clear API boundaries to support future feature expansion.

SwiftTrade - Day Trading System

Python, REST APIs, Docker, JMeter, MongoDB, RabbitMQ, Redis

- Designed and developed a full-stack day trading system supporting user authentication, wallet management, market orders, and limit orders.
- Implemented a custom FIFO-based order matching engine with sorted buy and sell order books and support for partial order fulfillment.
- Exposed secure REST APIs and containerized the system using Docker to enable scalable deployment and testing.

Procedural Building Generator

Unity, C#, Procedural Generation

- Developed a procedural building generation system in Unity using C#, enabling automatic creation of modular and scalable building layouts.
- Designed algorithms to dynamically assemble structures from reusable components, increasing level variety and design efficiency.
- Focused on performance optimization and editor usability to support rapid iteration and real-time generation.

Mini Football Game

Unreal Engine, Blueprints, Game Physics

- Built a 3D mini football game prototype in Unreal Engine, implementing player controls, ball physics, goal detection, and scoring logic.
- Designed interactive gameplay mechanics using Unreal Engine Blueprints to ensure responsive movement and realistic ball interaction.
- Implemented visual and gameplay feedback systems, including score updates and goal reactions, to enhance player experience.

RhythmBeats

Arduino, Teachable Machine, Sensors

- Developed a physical music composition system using Arduino and sensor-based inputs to detect and classify musical blocks.
- Integrated a Teachable Machine model to map physical inputs to musical notes and instrument sounds.
- Designed the system to provide reliable real-time feedback and an engaging tangible interaction experience.

SceneCraft

Arduino, AI, Physical Interaction

- Built an interactive storytelling tool that combines physical object manipulation with AI-generated narrative prompts.
- Designed intuitive tangible interactions to support creativity and engagement for young users.
- Integrated AI logic to dynamically generate story elements based on user input sequences.

Additional

Interests: Learning modern technologies, working on cars, collecting sneakers, video games